LINUX BASICS AND COMMAND

In this Project I will demonstrate some basic Linux commands and their uses with screenshot.

Also I will add some questions from tryhacckme which I solved.

So what Is Linux, Linux is nothing but operating system just like Windows and Maco's.

It has different versions just like windows and macOS.
Lets start with our few basics commands.

1.ECHO – Is the command that output txt we provide.

```
tryhackme@linux1:~$ echo welcom boy
welcom boy
tryhackme@linux1:~$
```

2.WHOAMI – This command tells us what user currently wee are logged in.see screenshot below.

```
tryhackme@linux1:~$ echo welcom boy
welcom boy
tryhackme@linux1:~$ whomi

Command 'whomi' not found, did you mean:

  command 'whom' from deb mailutils-mh (1:3.7-2.1)
  command 'whom' from deb mmh (0.4-2)
  command 'whom' from deb nmh (1.7.1-6)
  command 'whoami' from deb coreutils (8.30-3ubuntu2)

Try: apt install <deb name>

tryhackme@linux1:~$ whoami
tryhackme
tryhackme@linux1:~$
```

As per above screenshot we can observe that I entered wrong spelling so it got error, by this we learn that Linux platform CLI is case sensitive.

3. ls – This command list all files and folders available in the directory. In the screenshot below you can see 4 folders are available in current directory.

```
tryhackme@linux1:~$ echo welcom boy
welcom boy
tryhackme@linux1:~$ whomi

Command 'whomi' not found, did you mean:

  command 'whom' from deb mailutils-mh (1:3.7-2.1)
  command 'whom' from deb mmh (0.4-2)
  command 'whom' from deb nmh (1.7.1-6)
  command 'whoami' from deb coreutils (8.30-3ubuntu2)

Try: apt install <deb name>

tryhackme@linux1:~$ whoami
tryhackme
tryhackme@linux1:~$ ls
access.log  folder1  folder2  folder3  folder4
tryhackme@linux1:~$
```

4. cd- This is the command to change current directory to desired one. For ex screenshot below shows command “cd folder1” to entre in folder1 similarly for folder 2 it will be “cd folder2”.

```
tryhackme@linux1:~$ echo welcom boy
welcom boy
tryhackme@linux1:~$ whomi

Command 'whomi' not found, did you mean:

  command 'whom' from deb mailutils-mh (1:3.7-2.1)
  command 'whom' from deb mmh (0.4-2)
  command 'whom' from deb nmh (1.7.1-6)
  command 'whoami' from deb coreutils (8.30-3ubuntu2)

Try: apt install <deb name>

tryhackme@linux1:~$ whoami
tryhackme
tryhackme@linux1:~$ ls
access.log  folder1  folder2  folder3  folder4
tryhackme@linux1:~$ cd folder1
tryhackme@linux1:~/folder1$
```

5.use command “ls” to list things in current folder1.

```
tryhackme@linux1:~$ echo welcom boy
welcom boy
tryhackme@linux1:~$ whomi

Command 'whomi' not found, did you mean:

  command 'whom' from deb mailutils-mh (1:3.7-2.1)
  command 'whom' from deb mmh (0.4-2)
  command 'whom' from deb nmh (1.7.1-6)
  command 'whoami' from deb coreutils (8.30-3ubuntu2)

Try: apt install <deb name>

tryhackme@linux1:~$ whoami
tryhackme
tryhackme@linux1:~$ ls
access.log  folder1  folder2  folder3  folder4
tryhackme@linux1:~$ cd folder1
tryhackme@linux1:~/folder1$ ls
tryhackme@linux1:~/folder1$
```

6.cat- concatenate, this command is verry useful foe outputting the contents of file.not only for text. Screenshot below shows the “cat note.txt” shows what note file content that is “hello world”.

```
tryhackme@linux1://$ cd /
tryhackme@linux1:/$ cd
tryhackme@linux1:~$ ls
access.log  folder1  folder2  folder3  folder4
tryhackme@linux1:~$ cd folder2
tryhackme@linux1:~/folder2$ cd
tryhackme@linux1:~$ cd folder3
tryhackme@linux1:~/folder3$ cd
tryhackme@linux1:~$ cd folder4
tryhackme@linux1:~/folder4$ ls
note.txt
tryhackme@linux1:~/folder4$ cat note.txt
Hello World!
tryhackme@linux1:~/folder4$
```

7.PWD- command will show us our current working directory. Screenshot below shows us our directory is “/home/tryhackme/folder4”

```
tryhackme@linux1://$ cd /
tryhackme@linux1:/$ cd
tryhackme@linux1:~$ ls
access.log  folder1  folder2  folder3  folder4
tryhackme@linux1:~$ cd folder2
tryhackme@linux1:~/folder2$ cd
tryhackme@linux1:~$ cd folder3
tryhackme@linux1:~/folder3$ cd
tryhackme@linux1:~$ cd folder4
tryhackme@linux1:~/folder4$ ls
note.txt
tryhackme@linux1:~/folder4$ cat note.txt
Hello World!
tryhackme@linux1:~/folder4$ pwd
/home/tryhackme/folder4
tryhackme@linux1:~/folder4$
```

8.well after using cd and ls command I came to knew where is note.txt file but for now I don’t know or I want to save time by skipping searching in every file for note.txt for this purpose I will use command “find” as shown below.

This command will list only note.txt files located in that current directory.

```
tryhackme@linux1:~/folder4$ cat note.txt
Hello World!
tryhackme@linux1:~/folder4$ pwd
/home/tryhackme/folder4
tryhackme@linux1:~/folder4$ find -name note.txt
./note.txt
tryhackme@linux1:~/folder4$
```

9. Lets see o don't know the name of that txt file for that I will use command **"find -name *.txt"** this command will list all txt files located in that directory or folder.

```
tryhackme@linux1:~/folder4$ pwd
/home/tryhackme/folder4
tryhackme@linux1:~/folder4$ find -name note.txt
./note.txt
tryhackme@linux1:~/folder4$ find -name *.txt
./note.txt
tryhackme@linux1:~/folder4$
```

10.grep- The grep command allows us to search the contents of files for specific values that we are looking for. Look at question below for finding flag.

Use grep on "access.log" to find the flag that has a prefix of "THM". What is the flag? **Note:** The "access.log" file is located in the "/home/tryhackme/" directory.

THM{ACCESS} ✓ Correct Answer

to answer above question I used command "**grep THM access.log**"this will list all entries related to THM in access.log and we found the flag.

```
tryhackme@linux1:~$ find -name *.log
./access.log
tryhackme@linux1:~$ grep THM access.log
13.127.130.212 - - [04/May/2021:08:35:26 +0000] "GET THM{ACCESS} lang=en HTTP/1.1" 404 360 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64
) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/77.0.3865.120 Safari/537.36"
tryhackme@linux1:~$
```

11.mkdir- This command is used to create new directory. Also you can see that I have used operator "**&&**" and command "cd" to change directory.

```
tryhackme@linux1:~$ mkdir test && cd test
tryhackme@linux1:~/test$
```

Above sc shows that

mkdir test -create folder named test

&& - is the operator who run the next command when previous command gets successful.

Cd test- will Lauch me in directory test.

12. operator > - this is the output redirector operator means we take output of command and send it somewhere else. See screenshot below here “echo vrushabh is sent to folder tryhackme” when I cat tryhackme it show “vrushabh”.

```
tryhackme@linux1:~/test$ cd
tryhackme@linux1:~$ echo vrushabh > tryhackme
tryhackme@linux1:~$ cat tryhackme
vrushabh
tryhackme@linux1:~$
```

13.operator >> - This is the same as we discussed previously about “>”.just one difference is that this operator don’t over right to the previous content like “>” do.

See sc below it will show you first output “vrushabh”and second is “vrushabh purvat”.

```
tryhackme@linux1:~/test$ cd
tryhackme@linux1:~$ echo vrushabh > tryhackme
tryhackme@linux1:~$ cat tryhackme
vrushabh
tryhackme@linux1:~$ echo vrushabh purvat >> tryhackme
tryhackme@linux1:~$ cat tryhackme
vrushabh
vrushabh purvat
tryhackme@linux1:~$
```

14. connecting via SSH- SSH is secure shell protocol which works on cryptography and data we send from device is human readable which can be tampered to avoid this SSH converts All human readable data in encrypted format. Screenshot below shows how to connect SSH via attack box in tryhackme.

```
root@ip-10-49-117-36:~# ssh tryhackme@10.49.160.200
The authenticity of host '10.49.160.200 (10.49.160.200)' can't be established.
ED25519 key fingerprint is SHA256:hRxMYIa56KrhaFUzw818y2PQr9LIb+sQlBdODi05lDE.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.49.160.200' (ED25519) to the list of known hosts
tryhackme@10.49.160.200's password:
Permission denied, please try again.
tryhackme@10.49.160.200's password:
Welcome to Ubuntu 20.04.6 LTS (GNU/Linux 5.15.0-1075-aws x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Sun Mar 22 11:30:44 UTC 2026

  System load:  0.0               Processes:             102
  Usage of /:   29.0% of 9.62GB   Users logged in:       0
  Memory usage: 33%               IPv4 address for eth0: 10.49.160.200
  Swap usage:   0%
```

15.-a – This "-a" command will list all the hidden files in that directory or folder.

Remember all hidden file have dot at beginning of their name for ex "note.txt" is the normal file while ".note.txt" is the hidden file.

Screenshot below shows all hidden files in the folder.

```
tryhackme@linux2: ~
tryhackme@linux2: $ ls
important  myfile            unknown1
tryhackme@linux2: $ ls -a
    .bash_logout            important
    .bashrc       .profile  myfile     unknown1
tryhackme@linux2: $
```

16.ls –help – This option will show possible commands which I can use with ls.

```
tryhackme@linux2: $ ls --hlep
ls: unrecognized option '--hlep'
Try 'ls --help' for more information.
tryhackme@linux2: $ ls --help
Usage: ls [OPTION]... [FILE]...
List information about the FILEs (the current directory by default).
Sort entries alphabetically if none of -cftuvSUX nor --sort is specified.

Mandatory arguments to long options are mandatory for short options too.
  -a, --all                  do not ignore entries starting with .
  -A, --almost-all           do not list implied . and ..
      --author               with -l, print the author of each file
  -b, --escape               print C-style escapes for nongraphic characters
      --block-size=SIZE      with -l, scale sizes by SIZE when printing them;
                               e.g., '--block-size=M'; see SIZE format below
  -B, --ignore-backups       do not list implied entries ending with ~
  -c                         with -lt: sort by, and show, ctime (time of last
                               modification of file status information);
                               with -l: show ctime and sort by name;
                               otherwise: sort by ctime, newest first
  -C                         list entries by columns
      --color[=WHEN]         colorize the output; WHEN can be 'always' (default
                               if omitted), 'auto', or 'never'; more info below
  -d, --directory            list directories themselves, not their contents
```

17. touch- The touch command used to create new file as shown in screenshot below.

```
tryhackme@linux2: $ touch note
tryhackme@linux2: $ ls
important                myfile            note  unknown1
tryhackme@linux2: $ touch txt
tryhackme@linux2: $ ls
important                myfile            note  txt  unknown1
tryhackme@linux2: $ touch php
tryhackme@linux2: $ ls
important                myfile            note  php  txt  unknown1
tryhackme@linux2: $
```

18.rm- As discussed above touch is used to create new file “rm”is used to remove any file.

```
tryhackme@linux2: $ rm note
tryhackme@linux2: $ ls
important  myfile            php  txt  unknown1
tryhackme@linux2: $
```

19. mkdir- similarly “mkdir”command is used to create new directory and “r -R <dir>” used to remove directory.

20.ls -l -This command will list all content of the directory as shown below.

```
tryhackme@linux2: $ ls -l
total 28
-rw-rw-r-- 1 tryhackme tryhackme    0 Mar 22 11:52 2
-rw-rw-r-- 1 tryhackme tryhackme    0 Mar 22 11:52 3
-rw-rw-r-- 1 tryhackme tryhackme    8 Mar 22 11:59 file.txt
-rw-rw-r-- 1 tryhackme tryhackme    8 Mar 22 11:59 file2.txt
-rw-r--r-- 1 user2     user2       14 May  5  2021 important
-rw-r--r-- 1 tryhackme tryhackme   16 May  5  2021 myfile
drwxr-xr-x 2 tryhackme tryhackme 4096 May  4  2021
-rw-rw-r-- 1 tryhackme tryhackme    0 Mar 22 11:52 myfolder1
-rw-rw-r-- 1 tryhackme tryhackme    0 Mar 22 11:52 note
-rw-rw-r-- 1 tryhackme tryhackme    0 Mar 22 11:43 php
-rw-rw-r-- 1 tryhackme tryhackme    0 Mar 22 11:50 txt
-rw-r--r-- 1 tryhackme tryhackme   17 May  4  2021 unknown1
drwxrwxr-x 2 tryhackme tryhackme 4096 Mar 22 11:53
-rw-rw-r-- 1 tryhackme tryhackme    0 Mar 22 11:53 vrushabh3
tryhackme@linux2: $
```

21.su- for switching user we can use “su <user name>”.

22.nano – command “nano” is text editor which can do various tasks in linux.

- Searching for text
- Copying and Pasting
- Jumping to a line number
- Finding out what line number you are on.

Edit "task3" located in "tryhackme"'s home directory using Nano. What is the flag?

THM{TEXT_EDITORS} ✓ Correct Answer

```
  GNU nano 4.8                    task3
THM{TEXT_EDITORS}
```

As shown above I used nano editor to find the flag.

23.wget- To download file from http server “wget”command is used.foe ex lets assuem I know myfile.txt web adress in this case coomand will look like

“wget https://assets.tryhackme.com/additional/linux-fundamentals/part3/myfile.txt”.

24. serving as python3 server- The command below in screenshot shows **“python3 -m http.server”**used to serve It as server from where you can download files to another location. Using this command I found flag.

Download the file http://10.49.181.176:8000/.flag.txt (opens in new tab) onto the TryHackMe AttackBox. Remember, you will need to do this in a new terminal.

What are the contents?

THM{WGET_WEBSERVER} ✓ Correct Answer

```
tryhackme@linux3: $ wget http://10.49.181.176:8000/.flag.txt
--2026-03-22 12:53:33--  http://10.49.181.176:8000/.flag.txt
Connecting to 10.49.181.176:8000... failed: Connection refused.
tryhackme@linux3: $ python3 -m http.server
Serving HTTP on 0.0.0.0 port 8000 (http://0.0.0.0:8000/) ...
10.49.117.36 - - [22/Mar/2026 12:55:42] "GET /.flag.txt HTTP/1.1" 200 -
```

```
drwx------  3 root root 4096 Feb 25 12:55 .pki
-rw-r--r--  1 root root  189 Feb 25 12:19 .profile
-rw-------  1 root root   46 Mar 18 11:13 .python_history
-rw-r--r--  1 root root   66 Feb 25 11:39 .selected_editor
drwx------  2 root root 4096 Mar 22 12:39 .ssh
drwxr-xr-x  2 root root 4096 Feb 27 13:13 .sstimap
-rw-------  1 root root  799 Feb 25 11:35 .viminfo
drwxr-xr-x  2 root root 4096 Mar  3 09:29 .vnc
-rw-r--r--  1 root root  321 Feb 27 13:35 .wget-hsts
drwxr-xr-x  4 root root 4096 Feb 27 12:04 Desktop
drwxr-xr-x  2 root root 4096 Feb 25 11:07 Documents
drwxr-xr-x  2 root root 4096 Feb 25 11:07 Downloads
drwxr-xr-x  2 root root 4096 Feb 26 10:39 Pictures
drwxr-xr-x  3 root root 4096 Feb 26 10:51 Postman
drwxr-xr-x 42 root root 4096 Mar 18 11:25 Rooms
drwxr-xr-x  2 root root 4096 Mar 18 11:43 Scripts
drwxr-xr-x  3 root root 4096 Mar 18 10:48 go
drwx------  7 root root 4096 Mar 11 08:40 snap
root@ip-10-49-117-36:~# cat flag.txt
cat: flag.txt: No such file or directory
root@ip-10-49-117-36:~# cat .flag.txt
THM{WGET_WEBSERVER}
```

25.ps- This command shows running processes on your syatem.

```
tryhackme@linux3: $ ps
    PID TTY          TIME CMD
   1063 pts/0    00:00:00 bash
   1123 pts/0    00:00:00 ps
tryhackme@linux3: $
```

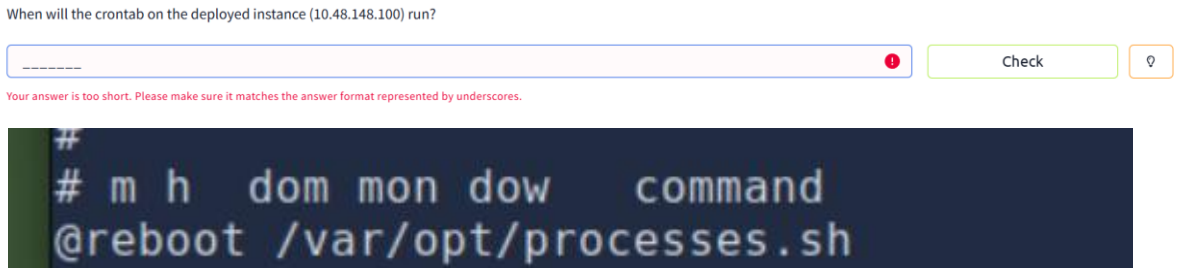
26.ps aux- To see the processes run by other users and those that don't run from a session (i.e. system processes), we need to provide **aux** to the ps command like so: ps aux.as shown below

```
tryhackme@linux3: $ ps aux
USER         PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root           1  0.1  0.6 103728 12512 ?        Ss   12:36   0:01 /sbin/init
root           2  0.0  0.0      0     0 ?        S    12:36   0:00 [kthreadd]
root           3  0.0  0.0      0     0 ?        I<   12:36   0:00 [rcu_gp]
root           4  0.0  0.0      0     0 ?        I<   12:36   0:00 [rcu_par_gp]
root           5  0.0  0.0      0     0 ?        I<   12:36   0:00 [slub_flushwq
root           6  0.0  0.0      0     0 ?        I<   12:36   0:00 [netns]
root           8  0.0  0.0      0     0 ?        I<   12:36   0:00 [kworker/0:0H
root          10  0.0  0.0      0     0 ?        I<   12:36   0:00 [mm_percpu_wq
root          11  0.0  0.0      0     0 ?        S    12:36   0:00 [rcu_tasks_ru
root          12  0.0  0.0      0     0 ?        S    12:36   0:00 [rcu_tasks_tr
root          13  0.0  0.0      0     0 ?        S    12:36   0:00 [ksoftirqd/0]
root          14  0.0  0.0      0     0 ?        I    12:36   0:00 [rcu_sched]
root          15  0.0  0.0      0     0 ?        S    12:36   0:00 [migration/0]
root          16  0.0  0.0      0     0 ?        S    12:36   0:00 [idle_inject/
root          18  0.0  0.0      0     0 ?        S    12:36   0:00 [cpuhp/0]
root          19  0.0  0.0      0     0 ?        S    12:36   0:00 [kdevtmpfs]
root          20  0.0  0.0      0     0 ?        I<   12:36   0:00 [inet_frag_wq
```

To background the process we can use “ctrl+z” similarly we an use “fg” to foreground the processes.

“Systemd” is the manager process that starts as soon as the machine boots, under this process every “child process” takes place.

27.crontab- This is the process that starts during boot, this is nothing but automated process which will execute on specified date and time. See screenshot below.



28. we have discussed about logs above but some important directories like logs are located in/

1./home/ - users personal file

2./root/ - root users home

3./etc/ - config files.

4./var/loga/ - logs as discussed above.

5./usr/bin/ - programs

6./bin/ essential program

7./tmp/- temporary files

8./dev/ - devices

9./proc/ - process info

10./boot. – boot files

11./opt/- third party apps.

So as per discussed above we have discussed some basic commands used in linux system.